

SDA Vision / Research report

infographic.pdf

Global Intelligence Summary

Information Epidemiology

Tracking the Digital Contagion of
Maternal Vaccine Misinformation

Synthesised from a 6-month, 15-country social media surveillance operation.



Assessment summary

Field	Recorded detail
Finding	Inconclusive
Recorded verdict	inconclusive
Confidence	low
Indicative rating	27/100
Rating basis	rounded mean of available frame scores; not a calibrated probability
Explanation	The sampled frames leave the assessment inconclusive. Review the frame results and source context.
Coverage	pdf - 4 frame(s) analysed of 15 found
Review context	Research assessment. Review alongside source information and context. When credentials are missing, origin remains unresolved by this check. Use these findings to inform a documented human review.

Provider overview

Provider	Type / status	Assessment	Rating
Claude Vision	vision / ok	synthetic likely	62/100
OpenAI Vision	vision / ok	inconclusive	15/100
Gemini Vision	vision / ok	authentic likely	30/100
C2PA Content Credentials	provenance / ok	inconclusive	n/a
Local forensic cues	forensic / ok	partially synthetic	41/100

Provider evidence / 1

Field	Recorded detail
Provider	Claude Vision
Model	claude-opus-5-5
Status	ok
Assessment	synthetic likely
Confidence	low
Role in assessment	Additional finding
Provider summary	Worst of 4 frames (figure p15): synthetic likelihood 62/100; tells: not a camera photo: flat digital graphic/slide, so photographic forensics (sensor noise, lens blur, lighting) do not apply, inconsistent centering of body text lines, with ragged, uneven offsets unlike typical layout-software alignment, polygonal network line-art with segments that end in mid-air, fade unevenly, and break into stray dashed fragments on the left, faint mottled texture in the flat background, atypical of vector-exported slides
Evidence 1	Frame figure p1: 40/100
Evidence 2	Frame figure p6: 45/100
Evidence 3	Frame figure p10: 45/100
Evidence 4	Frame figure p15: 62/100
Evidence 5	Not a camera photo: flat digital graphic/slide, so photographic forensics (sensor noise, lens blur, lighting) do not apply
Evidence 6	Inconsistent centering of body text lines, with ragged, uneven offsets unlike typical layout-software alignment
Evidence 7	Polygonal network line-art with segments that end in mid-air, fade unevenly, and break into stray dashed fragments on the left
Evidence 8	Faint mottled texture in the flat background, atypical of vector-exported slides
Evidence 9	1376x768 dimensions, which match a common native output size of some current image generators
Evidence 10	Heavy soft drop shadow on the headline, a stylistic pattern common in generated slide imagery
Evidence 11	Text is fully legible and correctly spelled, which weighs against older generators but not modern text-capable models
Field	Recorded detail
Evidence 12	Caveat: Designed graphics from a PDF can be produced by ordinary layout tools or by modern text-capable generators and look nearly identical. The layout

Field	Recorded detail
	irregularities and image dimensions are suggestive only, so check the source PDF metadata, its creation tool, and any C2PA/provenance data.

Provider evidence / 2

Field	Recorded detail
Provider	OpenAI Vision
Model	gpt-6-astra
Status	ok
Assessment	inconclusive
Confidence	low
Role in assessment	Additional finding
Provider summary	Worst of 4 frames (figure p6): synthetic likelihood 15/100; tells: Legible, coherent text with consistent repeated letterforms., Regular two-column layout with aligned rounded panels., Consistent bold labels and uniform bullet styling., No photographic base or identifiable inpainting boundaries.
Evidence 1	Frame figure p1: 12/100
Evidence 2	Frame figure p6: 15/100
Evidence 3	Frame figure p10: 12/100
Evidence 4	Frame figure p15: 15/100
Evidence 5	Legible, coherent text with consistent repeated letterforms.
Evidence 6	Regular two-column layout with aligned rounded panels.
Evidence 7	Consistent bold labels and uniform bullet styling.
Evidence 8	No photographic base or identifiable inpainting boundaries.
Evidence 9	Caveat: This is a designed infographic, not a camera photo; conventional design and AI-assisted production cannot reliably be distinguished from the rendered image alone.

Provider evidence / 3

Field	Recorded detail
Provider	Gemini Vision
Model	gemini-3.8-flash
Status	ok
Assessment	authentic likely
Confidence	medium
Role in assessment	Additional finding
Provider summary	Worst of 4 frames (figure p15): synthetic likelihood 30/100; tells: sharp vector-style geometric graphics, clean digital typography, standard graphic design layout artifacts
Evidence 1	Frame figure p1: 28/100
Evidence 2	Frame figure p6: 25/100
Evidence 3	Frame figure p10: 25/100
Evidence 4	Frame figure p15: 30/100
Evidence 5	Sharp vector-style geometric graphics
Evidence 6	Clean digital typography
Evidence 7	Standard graphic design layout artifacts
Evidence 8	Caveat: Standard presentation slide graphics can be synthesized or edited via automated tools.

Content Credentials / provenance

Field	Recorded detail
Provider	C2PA Content Credentials
Model	Not recorded
Status	ok
Assessment	inconclusive
Confidence	Not recorded
Role in assessment	Additional finding
Provider summary	Embedded credentials: none found. Origin remains unresolved by this check.
Evidence 1	Coverage: credentials embedded in the submitted file. External credentials were not retrieved.
Evidence 2	Verifier: c2pa-python 0.37.12; native SDK 0.91.0; specification target 2.4; policy embedded-offline-reviewed-trust-v3.
Evidence 3	Offline embedded-manifest verification; no network trust-list, manifest or revocation refresh.
Evidence 4	No reviewed local C2PA trust snapshot installed. Provenance overrides are disabled.

Provider evidence / 5

Field	Recorded detail
Provider	Local forensic cues
Model	Not recorded
Status	ok
Assessment	partially synthetic
Confidence	low
Role in assessment	Additional finding
Provider summary	Worst of 4 frames (figure p1): Heuristic synthetic likelihood 41/100 from metadata + pixel statistics.
Evidence 1	Frame figure p1: 41/100
Evidence 2	Frame figure p6: 41/100
Evidence 3	Frame figure p10: 41/100
Evidence 4	Frame figure p15: 41/100
Evidence 5	No camera EXIF (common in AI exports, screenshots, and re-saves).
Evidence 6	ELA mean 0.4, peak 21 (uniform high ELA can indicate generation/heavy edit).
Evidence 7	High-frequency noise std 14.12 (very low/very uniform noise is a synthesis cue).
Evidence 8	Spectral peakiness 19.49 (periodic frequency peaks can mark generated textures).

Per-frame results

Frame	Assessment	Rating / 100
figure p1	inconclusive	28
figure p6	inconclusive	25
figure p10	inconclusive	25
figure p15	inconclusive	30

Evidence supporting this verdict

Field	Recorded detail
Record	No entries recorded.

Additional findings

Field	Recorded detail
Claude Vision	Worst of 4 frames (figure p15): synthetic likelihood 62/100; tells: not a camera photo: flat digital graphic/slide, so photographic forensics (sensor noise, lens blur, lighting) do not apply, inconsistent centering of body text lines, with ragged, uneven offsets unlike typical layout-software alignment, polygonal network line-art with segments that end in mid-air, fade unevenly, and break into stray dashed fragments on the left, faint mottled texture in the flat background, atypical of vector-exported slides
OpenAI Vision	Worst of 4 frames (figure p6): synthetic likelihood 15/100; tells: Legible, coherent text with consistent repeated letterforms., Regular two-column layout with aligned rounded panels., Consistent bold labels and uniform bullet styling., No photographic base or identifiable inpainting boundaries.
Gemini Vision	Worst of 4 frames (figure p15): synthetic likelihood 30/100; tells: sharp vector-style geometric graphics, clean digital typography, standard graphic design layout artifacts
Local forensic cues	Supporting observation; considered alongside the model assessments.
C2PA Content Credentials	Embedded credentials: none found. Origin remains unresolved by this check.

Extracted text

Field	Recorded detail
Recorded text	Geographic Discourse Extremes The Sceptics (USA & Italy) • Primary Driver: Mistrust in institutions (CDC, FDA, Italy's ASL). • Core Theme: Extrapolating harm from contraindicated vaccines to recommended ones. • Vector Activity: High presence of bot-associated hashtags. The Advocates (India & UK) • Primary Driver: Proactive, visible public health campaigns. • Core Theme: Child protection and successful programme launches (e.g., India's Mission Indradhanush). • Vector Activity: Organic tagging of trusted bodies (Public Health England).

How this assessment was reached

Field	Recorded detail
1. frame_agreement	frames=4; synthetic_frames=0; authentic_frames=0; rating_spread=5
2. verdict	value=inconclusive; confidence=low; reason=The sampled frames leave the assessment inconclusive. Review the frame results and source context.

Cite this analysis

Field	Recorded detail
Software (Harvard)	Martin, S. (2026) SDA Vision: Synthetic-media Discourse Analysis (version 0.1.0) [Computer software]. Manchester Metropolitan University. Available at: https://sdavision.io/
Software creator	Dr Sam Martin, Smart Data Research UK (UKRI) Fellow, Manchester Metropolitan University (MMU). ORCID: https://orcid.org/0000-0002-4466-8374 .
Funding acknowledgement	This work was supported by Smart Data Research UK, a UKRI investment; Grant number UKRI4010.
Analysis record	File: infographic.pdf; media type: pdf; recorded at: 2026-09-26T22:31:05+00:00; models: claude-opus-5-5, gemini-3.8-flash, gpt-6-astra.

File and analysis record

Field	Recorded detail
File	infographic.pdf
Analysis ID	f46ae041-994d-44fa-997c-8e6f23dcff90
Generated	2026-09-26T22:31:05+00:00
Tool	SDA Vision 0.1.0
Models	claude-opus-5-5, gemini-3.8-flash, gpt-6-astra
Method version	pdf-preparation-2026-09-26.1
Analysis lens	general
Original SHA-256	02915f051b6eb2c9ef0c21f5bb20c4229ca21324bf20172f3867c32c299a5924
Ingestion note 1	PDF preparation: pdf-input-2026-09-26.1 (pypdf text/figures; PDFium page rendering).
Ingestion note 2	Extracted 4 embedded figure(s) of 15 found (figure mode).
Ingestion note 3	No selectable text layer - vision models read the page/figure images.
Ingestion note 4	Fresh provider replies acquired for eleven-examples-refresh-20260926-r1; report assembled locally from the saved responses without further API calls.
Ingestion note 5	Provider latency values are restored from the acquisition receipts (maximum sampled-frame latency per provider); raw replies and assessment results are unchanged by this bookkeeping step.